

PDF 8: JSP (Java Server Pages)

1. JSP stands for:

- a) Java Servlet Pages
- b) Java Standard Pages
- c) Java Server Pages
- d) JavaScript Pages

Answer: c

2. What file extension must JSP files have?

- a) .java
- b) .html
- c) .jspx
- d) .jsp

Answer: d

3. JSP is written as:

- a) Pure Java with no HTML
- b) Ordinary HTML with Java mixed in using special tags
- c) Pure HTML with no Java
- d) XML only

Answer: b

4. What happens when Tomcat first processes a JSP file?

- a) It serves it directly as HTML
- b) It translates the JSP into a Java Servlet, compiles it, and executes it
- c) It converts it to XML
- d) It runs it as a CGI script

Answer: b

5. Java code in a JSP file is enclosed in which tags?

- a) <java> ... </java>

- b) <?php ... ?>
- c) <% ... %>
- d) {{ ... }}

Answer: c

6. In JSP, <%= expression %> is equivalent to:

- a) System.out.println(expression)
- b) out.print(expression)
- c) return expression
- d) echo expression

Answer: b

7. Each call to a JSP servlet is executed in:

- a) A new servlet instance
- b) The main thread
- c) A new thread
- d) A new process

Answer: c

8. The HTML portion of a JSP file is referred to as:

- a) Static content
- b) Template text
- c) Script text
- d) View code

Answer: b

9. Since there is only one JSP object, using instance variables requires:

- a) Multiple JSP files
- b) Synchronization
- c) Session variables only
- d) No special handling

Answer: b

10. What is the main advantage of JSP over writing pure Servlets for HTML output?

- a) JSP runs faster
- b) JSP is written as HTML with embedded Java, making it easier for web designers
- c) JSP does not require a container
- d) JSP does not use Java

Answer: b

11. In JSP lifecycle, what is "Page Translation"?

- a) Converting JSP to XML
- b) Translating JSP expressions (`<%= %>`) into `out.print()` statements
- c) Loading the servlet class
- d) Initializing the servlet instance

Answer: b

12. What does `jspInit()` do in the JSP lifecycle?

- a) Destroys the JSP instance
- b) Handles each request
- c) Initializes the servlet instance
- d) Compiles the JSP

Answer: c

13. In a Servlet that acts as an MVC Controller, what does it do with results?

- a) Displays results directly in the console
- b) Passes results to a JSP (view) to render the HTML response
- c) Writes directly to a database
- d) Sends the results to another Servlet only

Answer: b

14. JSP files are processed by:

- a) The web browser
- b) Apache HTTP Server directly
- c) Tomcat (or any JEE web container)
- d) A JavaScript engine

Answer: c

15. In MVC using Servlets and JSP, what does the JSP use to retrieve data?

- a) Direct database calls
- b) HTTP GET requests
- c) Java Beans filled with data by the Servlet
- d) Static HTML files

Answer: c

16. How is a Servlet different from a JSP in terms of code style?

- a) A Servlet is HTML with embedded Java; JSP is pure Java
- b) A Servlet is Java with HTML embedded inside; JSP is HTML with Java embedded inside
- c) They are written identically
- d) JSP uses Python while Servlets use Java

Answer: b

17. What does `<%@ taglib %>` directive do in a JSP?

- a) Imports a Java class
- b) Declares the use of a custom or standard tag library
- c) Creates a new JSP session
- d) Sets the page encoding

Answer: b

18. The browser or client sees which output from a JSP?

- a) The raw JSP code
- b) The Java source code
- c) Only the resultant HTML

- d) The bytecode

Answer: c

19. JSP files must be placed:

- a) Anywhere on the server
- b) Only in the WEB-INF directory
- c) Within the web application directory accessible by Tomcat
- d) In the Java classpath

Answer: c

20. What is a key reason to prefer JSP over writing all output in Servlets?

- a) JSP is faster than Servlets
- b) Mixing lots of HTML inside `out.println()` in Servlets is messy; JSP lets HTML be written naturally
- c) JSP does not require compilation
- d) JSP supports more languages

Answer: b

21–30: *(Continuing with Session Tracking, JSTL, AJAX, JDBC, and EL — all PDF 9–15)*

PDF 9: Session Tracking

21. HTTP is a _____ protocol, which makes session tracking necessary:

- a) Reliable
- b) Stateful
- c) Stateless
- d) Persistent

Answer: c

22. Which of the following is NOT one of the three underlying session tracking mechanisms?

- a) Cookies
- b) URL Rewriting

- c) Hidden Form Fields
- d) SSL Certificates

Answer: d

23. What is a cookie?

- a) A server-side database record
- b) A small piece of information sent by a server to the browser, saved and sent back on future requests
- c) A JavaScript function
- d) An HTTP request method

Answer: b

24. What is the main disadvantage of using cookies for session tracking?

- a) They are too large
- b) Users can turn off cookies for privacy or security reasons
- c) They only work on mobile devices
- d) They cannot store session IDs

Answer: b

25. URL rewriting for session tracking adds the session ID as:

- a) A cookie
- b) A hidden form field
- c) An added parameter in the URL (e.g., ?sessionid=688)
- d) An HTTP header

Answer: c

26. What is the disadvantage of URL rewriting?

- a) It is not supported by browsers
- b) It only works with dynamically created documents (not static HTML)
- c) It requires cookies to be enabled
- d) It sends too much data

Answer: b

27. Hidden form fields for session tracking work by:

- a) Storing session data in the browser's local storage
- b) Embedding session data in hidden <input> fields submitted with forms
- c) Writing session data to a cookie
- d) Appending session data to the URL

Answer: b

28. What does the Servlet Session Tracking API provide over raw mechanisms?

- a) Only cookie support
- b) High-level programming APIs and a framework for managing sessions automatically
- c) Only URL rewriting support
- d) Direct database session storage

Answer: b

29. An online shopping cart is an example of why we need:

- a) JDBC
- b) Cookies only
- c) Session tracking to maintain state across multiple requests
- d) Static HTML

Answer: c

30. Hidden form fields break down when:

- a) The user has JavaScript enabled
- b) Used with static documents, bookmarked pages, or emailed documents
- c) The form has more than 10 fields
- d) The server uses HTTPS

Answer: b